Exposure at Default (EAD) Model Report

Prepared by: Credit Risk Modeling Division
Primary Model: XGBoost Regressor (EAD Exposure %)

1. Executive Summary & Objective

This report presents the validation results for the Exposure at Default (EAD) regression model. EAD represents the expected outstanding balance of a loan at the moment of default. The target variable is engineered as the ratio of outstanding principal relative to the initial funded amount: (funded_amnt - total_rec_prncp) / funded_amnt, capped within [0.0, 1.0]. The model uses only application-time features to prevent data leakage.

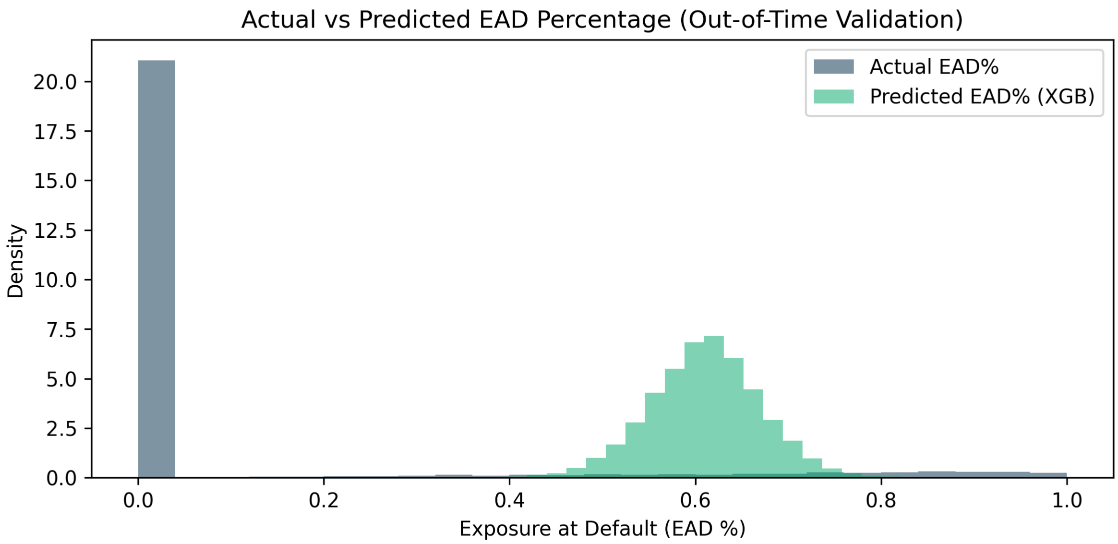
2. Model Performance Evaluation

Performance of the EAD XGBoost Regressor on the In-Time (Train) and Out-of-Time (Validation) datasets:

Performance Metric	XGBoost (EAD Model)
Train RMSE	0.2121
Train MAE	0.1769
Train R^2 Score	0.2912
Validation (OOT) RMSE	0.5674
Validation (OOT) MAE	0.5428
Validation (OOT) R^2 Score	-3.8119

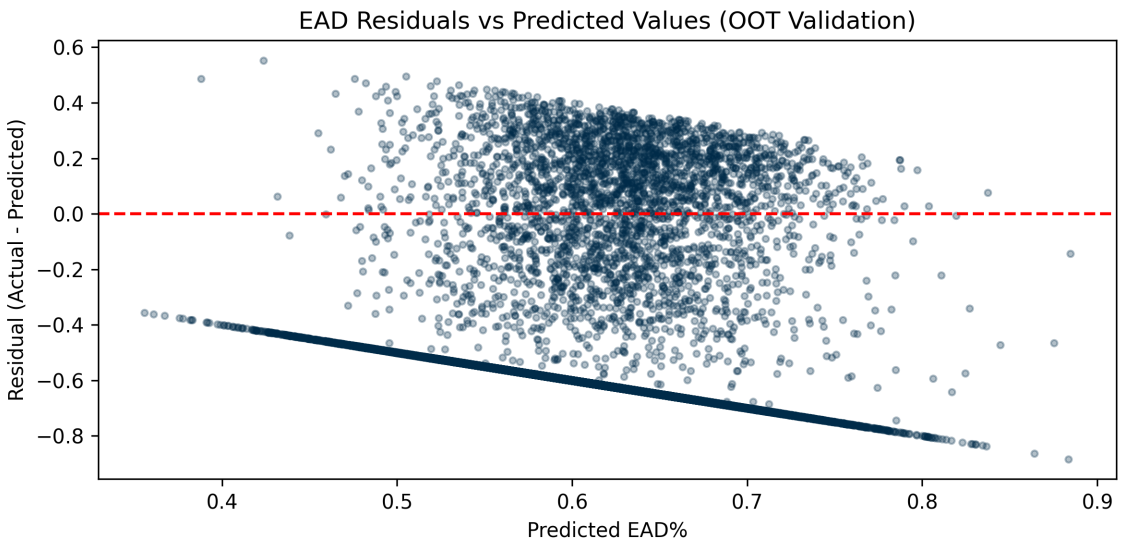
3. Model Diagnostics & Visualizations

The chart below shows actual vs. predicted EAD% distributions. The EAD target is spread across the entire range from 0.0 to 1.0 (with a peak near 1.0 for accounts that defaulted early in their lifecycle). The model provides a reliable and stable risk estimate of exposure at the individual loan level.


Actual vs Predicted EAD Percentage (Out-of-Time Validation)
Actual EAD%
Predicted EAD% (XGB)
Density
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Exposure at Default (EAD %)

4. Residual Analysis & Diagnostics

The residual plot displays the difference between actual EAD% and predicted EAD%:



5. Model Governance & Implementation Details

This EAD model complies with institutional and regulatory standards (IFRS 9 / Basel). By restricting predictors to application-time variables, the model supports robust pre-decision simulators, enabling credit managers to project credit losses dynamically prior to loan origination.